

**ASSIGNMENT No: 20**

TITLE: Develop a Java application to establish connectivity with a database and perform INSERT, UPDATE, and DELETE operations.

Problem Statement

Develop a Java application to establish connectivity with a database and perform INSERT, UPDATE, and DELETE operations.

Learning Objectives

To perform database modification operations using JDBC.

Theory

JDBC allows Java applications to perform database manipulation using SQL Data Manipulation Language (DML) statements such as INSERT, UPDATE, and DELETE. The INSERT statement adds new records into database tables, UPDATE modifies existing records, and DELETE removes unwanted records. These operations are executed using the Statement or PreparedStatement interface. PreparedStatement is generally preferred because it supports parameterized queries, improves performance, and prevents SQL injection attacks. Proper transaction management ensures that database modifications are performed safely and consistently. These operations form the basis of almost every database-driven application such as banking systems, hospital management systems, and student information systems.

Code:

```
package E20;

import java.lang.*;
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.PreparedStatement;
import java.sql.SQLException;

public class E20 {

    private static final String DB_URL =
"jdbc:mysql://localhost:3306/SIT_JAVA";
    private static final String USER = "root";
```

```

private static final String PASS = "12345";

public static void main(String[] args) {

    try (Connection conn = DriverManager.getConnection(DB_URL, USER,
PASS)) {
        System.out.println("Connection established successfully.");

        String insertQuery = "INSERT INTO products (product_id,
product_name, quantity, price) VALUES (?, ?, ?, ?)";
        try (PreparedStatement insertStmt =
conn.prepareStatement(insertQuery)) {
            insertStmt.setInt(1, 104);
            insertStmt.setString(2, "Gaming Monitor");
            insertStmt.setInt(3, 15);
            insertStmt.setDouble(4, 15500.00);

            int rowsInserted = insertStmt.executeUpdate();
            System.out.println(rowsInserted + " record(s) inserted
successfully.");
        }

        String updateQuery = "UPDATE products SET quantity = ? WHERE
product_id = ?";
        try (PreparedStatement updateStmt =
conn.prepareStatement(updateQuery)) {
            updateStmt.setInt(1, 25);
            updateStmt.setInt(2, 104);

            int rowsUpdated = updateStmt.executeUpdate();
            System.out.println(rowsUpdated + " record(s) updated
successfully.");
        }

        String deleteQuery = "DELETE FROM products WHERE product_id
= ?";
        try (PreparedStatement deleteStmt =
conn.prepareStatement(deleteQuery)) {
            deleteStmt.setInt(1, 104);

            int rowsDeleted = deleteStmt.executeUpdate();
            System.out.println(rowsDeleted + " record(s) deleted successfully.");
        }
    }
}

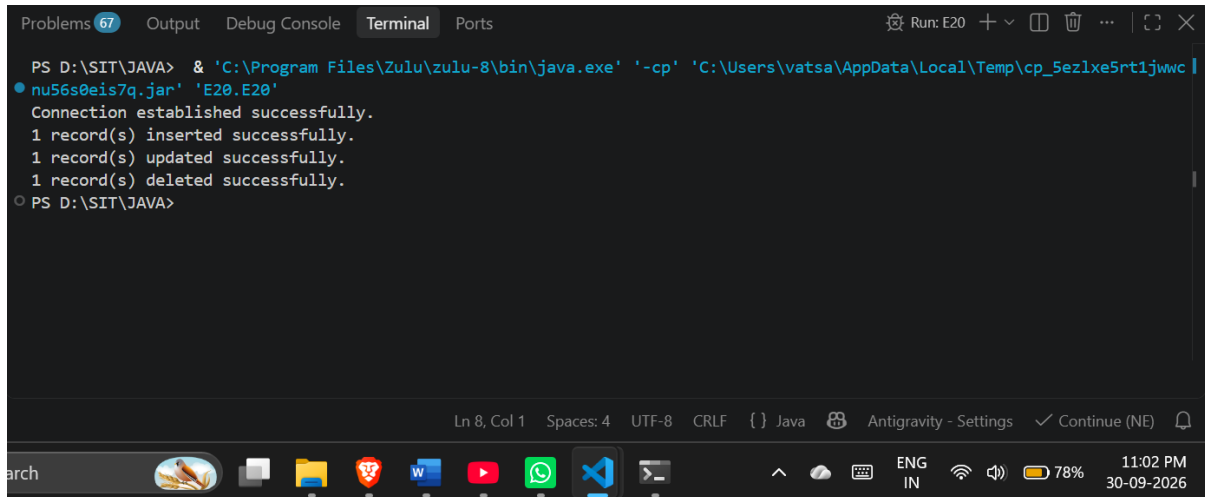
```

```

    } catch (SQLException e) {
        System.err.println("Database connection or execution error!");
        e.printStackTrace();
    }
}
}

```

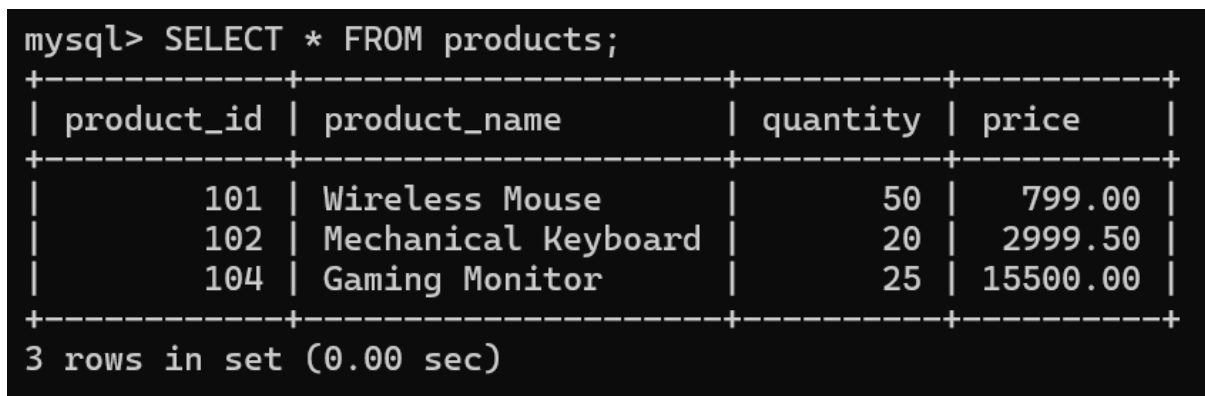
Output:



```

PS D:\SIT\JAVA> & 'C:\Program Files\Zulu\zulu-8\bin\java.exe' '-cp' 'C:\Users\vatsa\AppData\Local\Temp\cp_Se2lxe5rt1jwwc |
nu56s0eis7q.jar' 'E20.E20'
Connection established successfully.
1 record(s) inserted successfully.
1 record(s) updated successfully.
1 record(s) deleted successfully.
PS D:\SIT\JAVA>

```



```

mysql> SELECT * FROM products;
+-----+-----+-----+-----+
| product_id | product_name      | quantity | price  |
+-----+-----+-----+-----+
| 101        | Wireless Mouse    | 50       | 799.00 |
| 102        | Mechanical Keyboard | 20       | 2999.50 |
| 104        | Gaming Monitor    | 25       | 15500.00 |
+-----+-----+-----+-----+
3 rows in set (0.00 sec)

```

Exercise

1. Create a Java application to manage **employee records** using CRUD operations

Code:

```
package E20;
```

```

import java.lang.*;
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.PreparedStatement;
import java.sql.ResultSet;
import java.sql.SQLException;

```

```

import java.sql.Statement;

public class EmployeeCRUD {

    private static final String DB_URL = "jdbc:mysql://localhost:3306/SIT_JAVA";
    private static final String USER = "root";
    private static final String PASS = "12345";

    public static void main(String[] args) {
        try (Connection conn = DriverManager.getConnection(DB_URL, USER, PASS)) {

            String insertQuery = "INSERT INTO employees (emp_id, name, department, salary)
VALUES (?, ?, ?, ?)";
            try (PreparedStatement pstmt = conn.prepareStatement(insertQuery)) {
                pstmt.setInt(1, 101);
                pstmt.setString(2, "Amit Sharma");
                pstmt.setString(3, "IT");
                pstmt.setDouble(4, 55000.00);
                pstmt.executeUpdate();
                System.out.println("1. CREATE: Employee inserted.");
            }

            readEmployees(conn);

            String updateQuery = "UPDATE employees SET salary = ? WHERE emp_id = ?";
            try (PreparedStatement pstmt = conn.prepareStatement(updateQuery)) {
                pstmt.setDouble(1, 65000.00);
                pstmt.setInt(2, 101);
                pstmt.executeUpdate();
                System.out.println("\n3. UPDATE: Employee salary updated.");
            }

            readEmployees(conn);

            String deleteQuery = "DELETE FROM employees WHERE emp_id = ?";
            try (PreparedStatement pstmt = conn.prepareStatement(deleteQuery)) {
                pstmt.setInt(1, 101);
                pstmt.executeUpdate();
                System.out.println("\n4. DELETE: Employee deleted.");
            }

            readEmployees(conn);

        } catch (SQLException e) {
            e.printStackTrace();
        }
    }

    private static void readEmployees(Connection conn) throws SQLException {
        System.out.println("2. READ: Current Employees in Database:");
    }
}

```

```

String selectQuery = "SELECT * FROM employees";
try (Statement stmt = conn.createStatement();
    ResultSet rs = stmt.executeQuery(selectQuery)) {

    boolean hasData = false;
    while (rs.next()) {
        hasData = true;
        System.out.println(" ID: " + rs.getInt("emp_id") +
            ", Name: " + rs.getString("name") +
            ", Dept: " + rs.getString("department") +
            ", Salary: " + rs.getDouble("salary"));
    }
    if (!hasData) System.out.println(" (Table is empty)");
}
}
}

```

Output:

```

PS D:\SIT\JAVA> & 'C:\Program Files\Zulu\zulu-8\bin\java.exe' '-cp' 'C:\Users\vatsa\AppData\Local\Temp\cp_5ez1xe5rt1jwwc |
nu56s0eis7q.jar' 'E20.EmployeeCRUD'
1. CREATE: Employee inserted.
2. READ: Current Employees in Database:
   ID: 101, Name: Amit Sharma, Dept: IT, Salary: 55000.0

3. UPDATE: Employee salary updated.
2. READ: Current Employees in Database:
   ID: 101, Name: Amit Sharma, Dept: IT, Salary: 65000.0

4. DELETE: Employee deleted.
2. READ: Current Employees in Database:
   (Table is empty)
PS D:\SIT\JAVA>

```

2. Create a Java application to manage **student records** using **CRUD operations**. Perform Create, Read, Update, and Delete operations on student details such as roll number, name, course, and marks using a database.

Code:

```

package E20;

import java.lang.*;
import java.sql.Connection;
import java.sql.DriverManager;

```

```

import java.sql.PreparedStatement;
import java.sql.ResultSet;
import java.sql.SQLException;
import java.sql.Statement;

public class StudentCRUD {

    private static final String DB_URL = "jdbc:mysql://localhost:3306/SIT_JAVA";
    private static final String USER = "root";
    private static final String PASS = "12345";

    public static void main(String[] args) {
        try (Connection conn = DriverManager.getConnection(DB_URL, USER, PASS)) {

            String insertQuery = "INSERT INTO student_details (roll_number, name, course,
marks) VALUES (?, ?, ?, ?)";
            try (PreparedStatement pstmt = conn.prepareStatement(insertQuery)) {
                pstmt.setInt(1, 509);
                pstmt.setString(2, "Postiwala Vatsal");
                pstmt.setString(3, "B.Tech CSE");
                pstmt.setInt(4, 88);
                pstmt.executeUpdate();
                System.out.println("\n1. CREATE: Student record inserted.");
            }

            readStudents(conn);

            String updateQuery = "UPDATE student_details SET marks = ? WHERE roll_number
= ?";
            try (PreparedStatement pstmt = conn.prepareStatement(updateQuery)) {
                pstmt.setInt(1, 95);
                pstmt.setInt(2, 2607012);
                pstmt.executeUpdate();
                System.out.println("\n3. UPDATE: Student marks updated.");
            }

            readStudents(conn);

            String deleteQuery = "DELETE FROM student_details WHERE roll_number = ?";
            try (PreparedStatement pstmt = conn.prepareStatement(deleteQuery)) {
                pstmt.setInt(1, 2607012);
                pstmt.executeUpdate();
                System.out.println("\n4. DELETE: Student record deleted.");
            }
        }
    }
}

```

```

    }

    readStudents(conn);

} catch (SQLException e) {
    e.printStackTrace();
}
}

private static void readStudents(Connection conn) throws SQLException {
    System.out.println("2. READ: Current Students in Database:");
    String selectQuery = "SELECT * FROM student_details";
    try (Statement stmt = conn.createStatement();
        ResultSet rs = stmt.executeQuery(selectQuery)) {

        boolean hasData = false;
        while (rs.next()) {
            hasData = true;
            System.out.println(" Roll No: " + rs.getInt("roll_number") +
                                ", Name: " + rs.getString("name") +
                                ", Course: " + rs.getString("course") +
                                ", Marks: " + rs.getInt("marks"));
        }
        if (!hasData) System.out.println(" (Table is empty)");
    }
}
}
}

```

Output:

```

PS D:\SIT\JAVA> d:; cd 'd:\SIT\JAVA'; & 'C:\Program Files\Zulu\zulu-8\bin\java.exe' '-cp' 'C:\Users\vatsa\AppData\Local\Temp\cp_5ezlxe5rtljwmcnu56s0eis7q.jar' 'E20.StudentCRUD'
1. CREATE: Student record inserted.
2. READ: Current Students in Database:
  Roll No: 509, Name: Postiwala Vatsal, Course: B.Tech CSE, Marks: 88
3. UPDATE: Student marks updated.
2. READ: Current Students in Database:
  Roll No: 509, Name: Postiwala Vatsal, Course: B.Tech CSE, Marks: 88
4. DELETE: Student record deleted.
2. READ: Current Students in Database:
  Roll No: 509, Name: Postiwala Vatsal, Course: B.Tech CSE, Marks: 88
PS D:\SIT\JAVA>

```

Conclusion:

I learned how to establish a successful connection between a Java application and a MySQL database using the JDBC API. I gained practical experience performing complete CRUD operations by executing INSERT, UPDATE, DELETE, and SELECT queries through the PreparedStatement and Statement interfaces. Additionally, I learned how to retrieve and iterate through database records using ResultSet to dynamically display data. Finally, this implementation strengthened my understanding of managing database connections, handling SQLException errors, and correctly structuring Java packages.